Problem 1. Mean Molecular Weights

Calculate the mean molecular weight per free electron μ_e and the mean molecular weight per particle μ for a fully ionized gas of pure helium, assuming that the helium is composed of equal numbers of helium nuclei and electrons.

Solution. $\mu_e = 2$, $\mu = \frac{4}{3}$ **Hint:** Consider the number of helium nuclei and electrons per unit mass.

Let n_{He} denote the number of helium nuclei per unit mass. Since the helium is composed of equal numbers of helium nuclei and electrons, the number of electrons per unit mass is also n_{He}. Therefore, the total number of particles per unit mass is $n_{He} + n_{He} = 2n_{He}$.

Derivation: Mean Molecular Weight per Electron μ_e

The mean molecular weight per free electron μ_e is defined as:

$$\mu_e = \frac{\text{Total mass per unit volume}}{\text{Total number of free electrons per unit volume}}$$

Since the helium is composed of equal numbers of helium nuclei and electrons, the total number of free electrons per unit mass is:

$$\text{Total number of free electrons per unit volume} = n_{He}$$

Assuming the helium nuclei are fully ionized, each nucleus contributes two electrons, so the total mass per unit volume is:

$$\text{Total mass per unit volume} = 4 n_{He} m_u$$

Substituting these into the definition of μ_e:

$$\mu_e = \frac{4 n_{He} m_u}{n_{He}} = 4 m_u$$

Since m_u is the atomic mass unit, we have:

$$\mu_e = 2$$

Derivation: Mean Molecular Weight per Particle μ

The mean molecular weight per particle μ is defined as:

$$\mu = \frac{\text{Total mass per unit volume}}{\text{Total number of particles per unit volume}}$$

Since the helium is composed of equal numbers of helium nuclei and electrons, the total number of particles per unit mass is:

$$\text{Total number of particles per unit volume} = 2 n_{He}$$

Substituting these into the definition of μ:

$$\mu = \frac{4 n_{He} m_u}{2 n_{He}} = 2 m_u$$

Since m_u is the atomic mass unit, we have:

$$\mu = \frac{4}{3}$$

Discussion

- The calculation correctly uses the number of helium nuclei per unit mass.
- The mean molecular weight per free electron μ_e is consistent with the expected value for fully ionized helium.
- The mean molecular weight per particle μ is also consistent with the expected value for fully ionized helium.
- The calculation assumes that the helium is fully ionized, which is a reasonable assumption for most astrophysical applications.
- The calculation does not account for any deviations from ideal gas behavior or any other complications that might arise in real astrophysical systems.

Problem 2. Equation of State with Gas + Radiation Pressure

Consider a gas made up of pure helium. Assume that the helium is completely ionized. Let $\mu_e = 2$ and $\mu = \frac{4}{3}$ (see Problem 1). Assume that the gas is composed of a mixture of ideal gas and radiation.

Hint: $\mu_e = 2$, $\mu = \frac{4}{3}$